



# Seaglider Ballasting Procedure

Before a sea trial, a Seaglider needs enough lead trimmed out (or in) that its maximum internal volume ( `volmax` ) gives the right thrust in the density of the water it's actually deploying into. This page covers the tank procedure for estimating `volmax` and turning it into a weight change. Once the glider is in the water, trim is refined dynamically from flight data – see [Trim & Flight Model](#).

## Source

Paraphrased from the APL-UW IOP office-hours session on ballasting and `volmax` estimation (June 2026) and the APL-UW IOP *SGX Documentation*. The tank method is deliberately simple – IOP's own philosophy is to nail down `volmax` in the tank and let the sea trial work out pitch/roll trim dynamically, rather than trying to model static centers precisely on paper. The neutral-point tank test and leash-check sections draw on operator service correspondence (2018–2023). Numbers quoted here are IOP starting points – confirm against the [Parameter Reference Manual](#) and defer to APL-UW IOP guidance.

## Equipment

- A freshwater tank large enough to fully submerge the glider **vertically** (deep enough that it can hang without touching bottom or breaking the surface). Around **2 m** is the practical minimum – that is what the glider needs to take up its nose-down surface attitude (roughly 60–70°) clear of the floor. A 3–4 m tank allows short "dives", but they tell you little that the static test doesn't. Shallow above-ground pools (~1.4 m) have been used, but they limit what you can check.
- A hanging scale or load cell suspended over the tank, to weigh the glider in water.
- A line to suspend the glider from – tied off at the rudder, since it hangs vertically.
- The glider's comms cable, connected and slack – you need a live link to read/set VBD position while the glider hangs in the tank. Leave the antenna disconnected and simply let it dangle; nothing should be expressing at the surface.